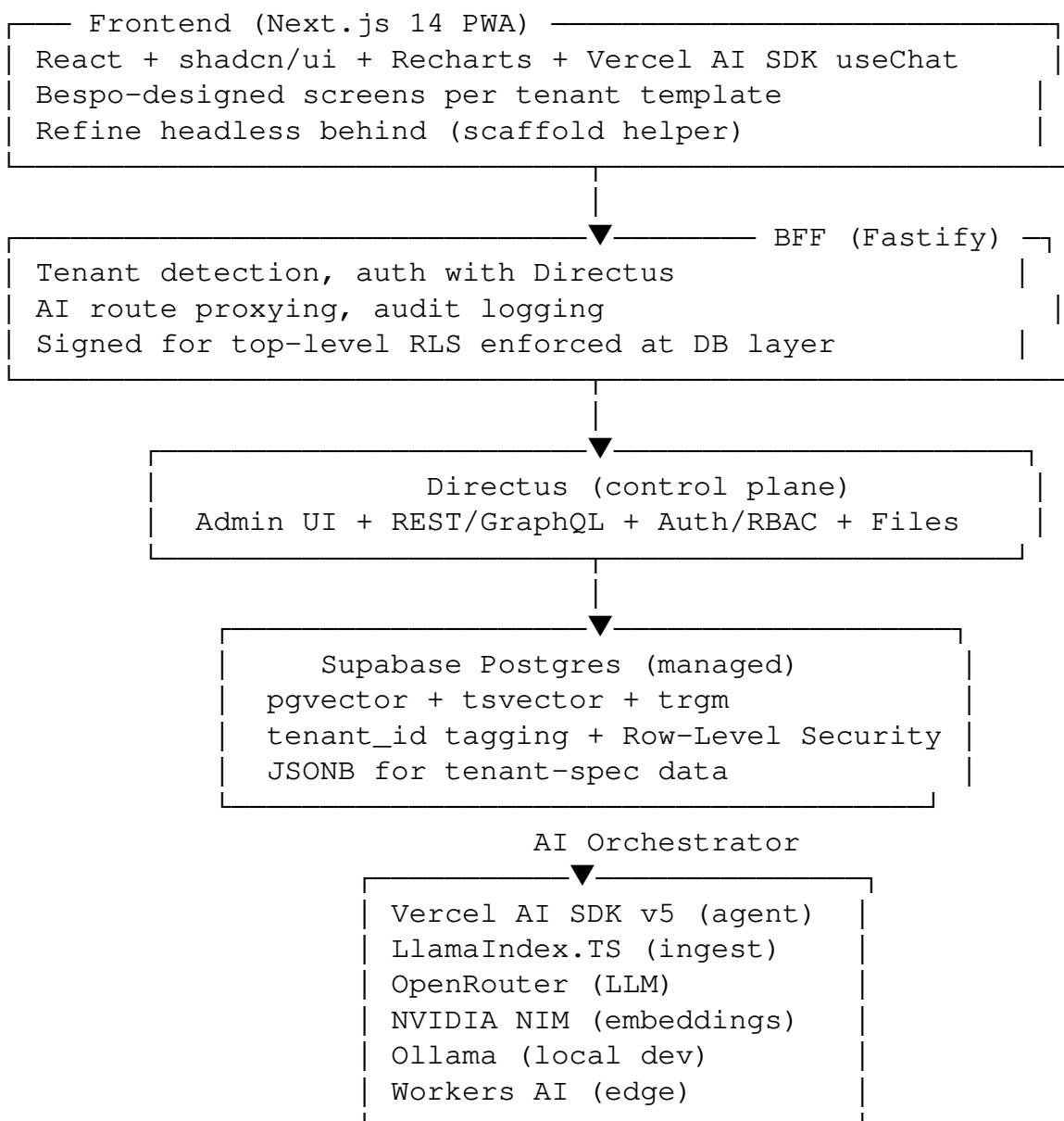


RubberTrack Multi-Tenant CRM/CMS/ Dashboard Build Handbook

Approved decisions: Tenancy = RLS + tiered escalation · AI = Vercel AI SDK v5 + LlamaIndex.TS + OpenRouter/NIM/Ollama/Workers · Control plane = Directus 12 · Templates = RubberTrack + generic Services

1. SYSTEM OVERVIEW

A multi-tenant business platform sold to multiple clients. Tenant (e.g., RubberTrack) is a template-driven deployment on shared infra with database-enforced isolation.



2. STACK & PINS

Layer	Selected	Version/pin	Purpose
Managed Postgres	Supabase (openalternative)	v17 +	pgvector, tsvector, trgm ready
Backend/API/ Admin/UI/ RBAC/Auth	Directus (openalternative)	directus:12.x	control plane
Frontend	Next.js 14 + React 18 + shadcn/ui + Recharts + Refine headless	next@14, shadcn, recharts, @refinedev/core@5	Figma screens, dashboard
AI orchestration	Vercel AI SDK v5 (own research, own research)	ai@5.x	agentic chat/ planner/tools
Ingestion & retriev	LlamaIndex.TS	llamaindex@latest	RAG/doc pipeline
Providers	OpenRouter primary (LLM), NVIDIA NIM (embeddings), Ollama (dev), Cloudflare Workers (edge)	bound by tenant	model routing

3. TENANT ONBOARDING (OPERATIONS)

1. Clone template (RubberTrack or Services).
2. Run script → tenant row + roles + permission presets + collections + seed.
3. Configure theme/labels/roles per tenant.
4. Tier A default (RLS). Tier B/C script when heavy/regulated.

4. TEMPLATES DEFINE WHAT TURNS ON

Module	Kind	RubberTrack	Services
Dashboard	su	✓	✓
Records	su	✓	✓
Parties (Customers/Suppliers)	su	✓	✓
Issues/Tickets	su	✓	✓
Attendance	su	✓	optional
News & Updates (CMS)	su	✓	✓
Doc Checker (AI)	su	✓	✓
Diff Checker	su	✓	✓
Checklists	su	✓	✓
Pending Documents	su	✓	✓
AI Assistant	su	✓	✓

5. PHASED BUILD PLAN (with exit criteria per phase)

Phase 0 — Scaffold (1 week)

- Supabase project + extension pgvector
- docker-compose: directus + frontend + bff + ai service + mock auth
- tenancy helper SQL `current_tenant()` + RLS policies on every entity
- Onboarding script → clone template, statuses, roles, permission presets
- roles: `privileged-admin`, `tenant-admin`, `staff-*`, `customer`

Phase 1 — Template Engine (2 weeks)

- Dynamic module registry (enabled per tenant template)
- RLS-aware Directus SDK hooks for custom screens
- RubberTrack demo template reproducing ALL Figma screens
- Import/Export Excel, screen-config editor (choose charts/cards)
- Generic Services template optional in parallel

Phase 2 — Dashboard + Hybrid Search (2 weeks)

- KPI engine (compute from DB, watch YoY)
- Chart builder with config-in-DB
- pgvector + tsvector + trgm hybrid search w/ merge + re-rank
- Global Search UI

Phase 3 — AI Platform (3 weeks)

- Provider router (OpenRouter/NIM/Ollama) w/ tenant config
- Assistant: planner→tools→synthesis, streaming UI
- Insights generator (nightly + on-demand)
- Doc Checker (extraction), Diff (side-by-side + flag mismatches)
- AI usage logs (costs/tools per tenant)

Phase 4 — Ops/Tenancy Escalation (2 weeks)

- Tenant-onboarding automation & cloning
- Schema-per-tenant / DB-per-tenant escalation script
- Per-tenant logical backup
- Isolation tests in CI

Phase 5 — White-label + release (2 weeks)

- per-tenant branding (theme.json in DB)
- External customer portal (login→scoped views)
- Deployment runbooks
- Demo-reset pipeline

6. SAMPLE TEMPLATE SCHEMA (RubberTrack)

Collections (in tenant template JSON):

- `records` — order/master data. eg `order_id`, `date`, `customer`, `supplier`, `grade`, `mt`, `fcl`, `price_usd`, `status`, `tenant_id`, JSONB extra
- `parties` — type `psvod/customer`; contact info, tags, embedding of details for assistant

- tickets — category, status, description, fields for kind like quality/shipment/doc
- hr_events — employee, department, attendance fields
- feed_items — category, title, description, priority
- checklists — customer/supplier-specific checklist sets
- files — uploads and associations
- embeddings — unified schema per Cerebras blog (source_type, source_id, vector, metadata, tenant_id)

Labels provided: records→Order Records, parties_customer→Customers, parties_supplier→Suppliers.

7. TENANT ISOLATION TESTS (proof)

- Test A: 2 tenants attempting read/write cross — expect deny.
- Test B: tenant-admin sees own collections only.
- Test C: staff cannot use Directus admin UI.
- Test D: external customer only sees own company rows.
- Test E: AI tools must inject tenant scope.

8. SECURITY RAC MATRIX

Role	Rest	Admin UI	Scope
privileged-admin	all layers	Full	system + templates
tenant-admin	within tenant	Full within tenant	roles + data
staff-sales	within tenant	No	sales records
staff-logistics	within tenant	No	ops records
staff-documentation	within tenant	No	docs
staff-technical	within tenant	No	tickets
customer	within tenant and own company	No	own documents

9. APPENDIX — Where each file lives post-scaffold

```
workspace/project/
├─ docker-compose.yml
├─ .env.example
├─ docs/
│   └─ build_handbook.md
├─ apps/
│   ├─ web/      ← Next.js + Refine headless + shadcn/ui
│   ├─ bff/      ← Fastify, tenant-detection + AI-proxy
│   └─ ai/       ← Vercel AI SDK + LlamaIndex + provider router
├─ infra/
```

```
|   |— tenancy/helper.sql      ← pg function + RLS
|   |— templates/rubbertrack.json
|   |— scripts/normalize.sh
|— README.md
```

END OF HANDBOOK

Signatories: celebrate under (privileged-admin) & project sponsor — both approvals on the 4 decisions recorded.